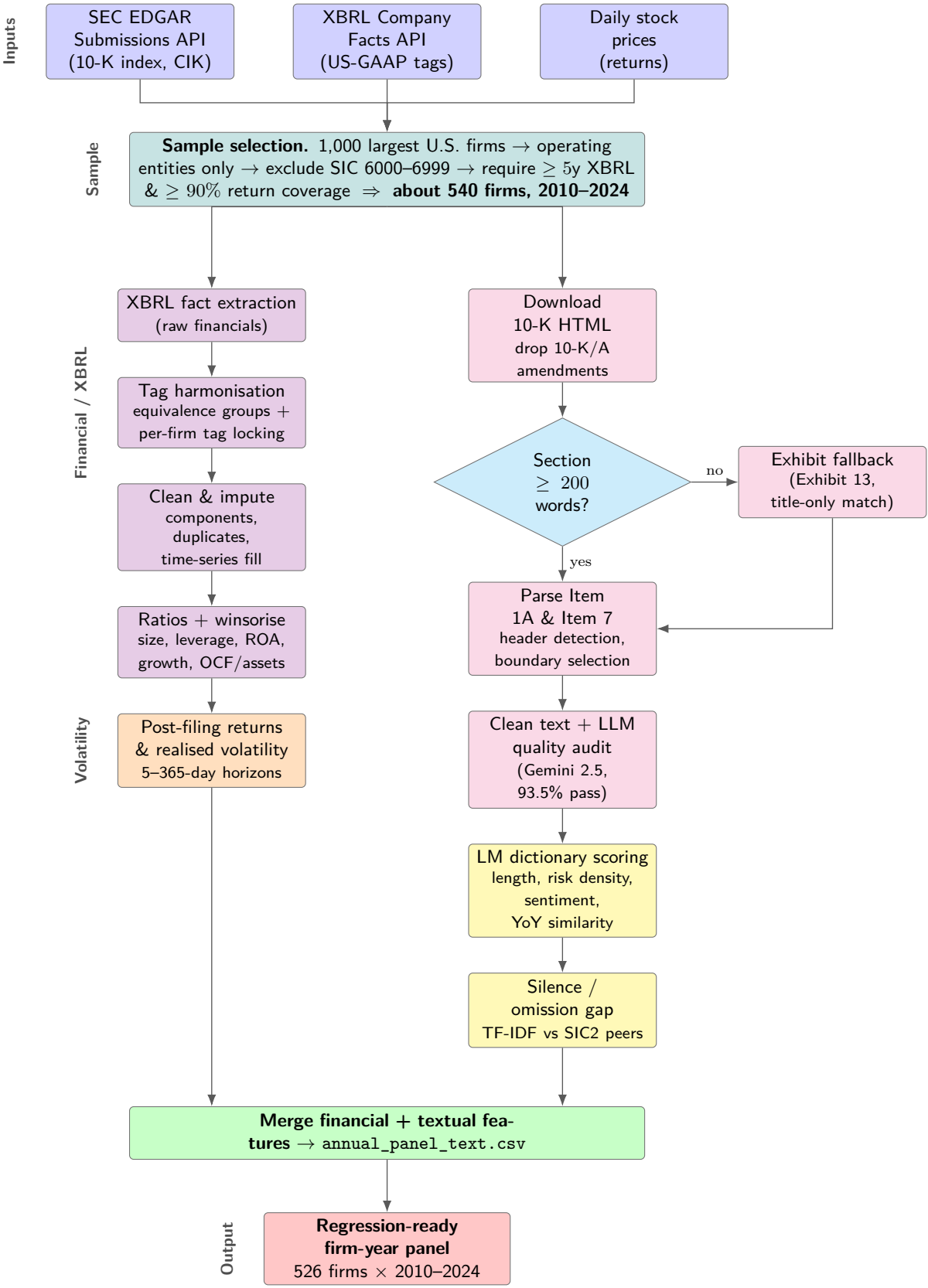

Inputs
SEC EDGAR Submissions API (10-K index, CIK)
XBRL Company Facts API (US-GAAP tags)
Daily stock prices (returns)
Sample
Sample selection. 1,000 largest U.S. firms → operating entities only → exclude SIC 6000–6999 → require ≥ 5y XBRL & ≥ 90% return coverage ⇒ about 540 firms, 2010–2024
Financial / XBRL
XBRL fact extraction (raw financials)
Tag harmonisation equivalence groups + per-firm tag locking
Clean & impute components, duplicates, time-series fill
Ratios + winsorise size, leverage, ROA, growth, OCF/assets
Download 10-K HTML drop 10-K/A amendments
Section ≥ 200 words?
no
Exhibit fallback (Exhibit 13, title-only match)
yes
Parse Item 1A & Item 7 header detection, boundary selection
Clean text + LLM quality audit (Gemini 2.5, 93.5% pass)
LM dictionary scoring length, risk density, sentiment, YoY similarity
Silence / omission gap TF-IDF vs SIC2 peers
Volatility
Post-filing returns & realised volatility 5–365-day horizons
Merge financial + textual features → annual_panel_text.csv
Output
Regression-ready firm-year panel 526 firms × 2010–2024